

Compiler/language-infrastructure engineer with professional LLVM and static-analysis experience. In UniversalToolchain I built dependency/conflict resolution, component/provider selection, deterministic pass ordering, typed artifact routing, and pre-execution plan checks; an architecture talk based on the project was accepted for LangDev'26.

2022: Public compiler project · 2023: Language tooling · 2025: Client/product + toolchain · 2026 →: Professional compilers / analysis

EXPERIENCE	SKILLS
2026 — present ISP RAS — Static Analysis Engineer Contribute to SharpChecker, ISP RAS's C#/.NET static-analysis platform.	Compiler Infrastructure dependency/conflict resolution, component/provider selection, deterministic pass ordering, typed artifact routing, pre-execution checks
July — August 2026 MCST — Compiler Engineering Intern - Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety. - Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.	Compiler / IR C++23, LLVM 22/IR, CFG/SSA, loop and interprocedural analysis, transformation applicability
2025 — confirmed Client development — Diploma.school Worked on a real product: jQuery dropdown interactions, responsive mobile/desktop navigation, and UI regression fixes. The work is confirmed no later than January 2025.	Backends / Runtime .NET, interpreter/CIL parity, SysV x86-64, register allocation, runtime composition Verification regression suites, LLVM Verifier, interpreter/backend parity, differential execution
PROJECTS	EDUCATION
UniversalToolchain Language features resolve dependencies and conflicts, select capability/runtime providers, obtain deterministic contribution ordering, and build typed artifact routes before runtime materialization. Current manifest: 1,324/1,324 tests; dedicated checks cover interpreter/CIL parity and configuration integrity before execution. LLVM Interprocedural Global IV Optimization An LLVM 22 pass localizes supported induction-like globals after interprocedural effect analysis, affine evolution, and a separate applicability check; APInt models fixed-width wraparound. 29 positive/negative regression cases cover LLVM Verifier checks, structural expectations, idempotence, and observable before/after behavior. x86-64 Compiler Backend & Register Allocation SSA/CFG validation → liveness/interference → register allocation/spills → phi lowering → SysV x86-64 emission. An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves.	HSE University — Software Engineering, 2026–2030
	RECOGNITION
	LangDev'26 — Build the Language, Then Make the Abstractions Disappear Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga. UniversalToolchain — Baltic Science and Engineering Competition First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain. MEPhI Junior Overall winner; First Degree Diploma, 96/100 for UniversalToolchain. Moscow / remote